

CODE :-

```
import numpy as np
p=np.array([[1,-1,-1,1],
            [-1,1,-1,1],
            [-1,1,1,-1],
            [1,-1,1,-1]])

w=np.zeros((4,4))
w=np.dot(p.T, p)

np.fill_diagonal(w,0)
print("Weight matrix:",w)

n=np.array([[1,1,-1,1],
            [-1,-1,-1,1],
            [1,1,1,-1],
            [1,-1,1,1]])
x=np.dot(w,n)
y= np.where(x>=0,1,-1)
print("Output pattern:\n", y)
```

OUTPUT :-

```
Weight matrix: [[ 0 -4  0  0]
 [-4  0  0  0]
 [ 0  0  0 -4]
 [ 0  0 -4  0]]
```

```
Output pattern:
[[ 1  1  1 -1]
 [-1 -1  1 -1]
 [-1  1 -1 -1]
 [-1 -1 -1  1]]
```